



Bioaccumulation of organic and inorganic contaminants in biota: A long-term evaluation in the Belgian part of the North Sea

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ABSTRACT

Determining the extent of pollution in the marine environment remains challenging. Polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenyls (PCBs), and metals can, during dredging operations, be transported from a port or harbour into the open sea, where they may exert a harmful effect on the marine ecosystem. To fully understand the impact of these chemicals, monitoring programs should not only focus on sediment concentrations, but should also take into account the bioaccumulated concentration in the tissue of multiple target organisms.

In this study, the concentration of primary contaminants is determined in common starfish (*Asterias rubens*), flying crab (*Liocarcinus holsatus*), and brown shrimp (*Crangon crangon*) and the difference in the concentration of contamination between different dredge disposal sites at open sea is investigated. Different factors such as lipid weight, dry weight, grain size, and total organic carbon were measured and used to understand the difference between the observed bioaccumulation and the measured sediment concentrations.

Key message: Different contaminants are detected in biota such as common starfish, flying crab and brown shrimp. These contaminants can be linked to dredging activities, with disposal sites associated with industrial ports showing higher contamination.

1. Introduction

The Belgian part of the North Sea covers an area of 3454 km² in which a broad variety of activities take place, such as the disposal of dredged material, fisheries, nature preservation, energy production through offshore wind farms, sand extraction and shipping. Some of these activities can result in the release of contaminants in the marine environment. Dredging is the removal of excessive sludge (dredged material) to guarantee the navigability of shipping channels and access to ports and harbours. Dredged material originating from ports, harbours or coastal areas can contain elevated levels of contaminants such as polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenyls (PCBs), and metals that may harm the marine environment (Dai et al., 2022).

PAHs can be released during oil refining, waste incineration, industrial and agricultural outfalls, fuel spillage and atmospheric deposition (Zychowski and Godard-Codding, 2017; Eisler, 1987). PCBs were used as coolant, plasticisers and lubricants, and were banned in the mid-'80s (Faroon et al., 2003). Therefore, current PCB concentrations in the marine environment are mainly historical. Nevertheless, the use of old equipment and the disposal of contaminated sediments can still result in contamination with PCBs (OSPAR commission, 2014; Warmuth and Ohno, 2013; Yuan et al., 2015). The contamination of metals can have a wide variety of anthropogenic sources such as mining, smelting ore, the incineration of fossil fuels, the (inappropriate) use of pesticides, etc. (Gürses et al., 2021; Paz et al., 2018; Rubio et al., 2017; Defarge et al., 2018). Additionally, PAHs and metals can, to a lower extent, also be released during natural processes such as forest fires and volcanic

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activities (Zychowski and Godard-Coddling, 2017), sea-salt spraying, biogenic activities and rock weathering (Masindi and Muedi, 2018).

PAHs, PCBs and metals can be released into the marine environment where they can exert a negative effect on the marine ecosystem. The EU aims to protect and conserve the health of our oceans and coasts through the Marine Strategy Framework Directive (MSFD). To achieve a Good Environmental Status (GES) of EU marine waters and to protect the marine resources and marine-related economic and social activities, eleven descriptors were proposed (European Commission, 2023). Descriptor 8 specifically envisages to protect the marine ecosystem against pollution effects, including monitoring criteria to evaluate the overall contamination and compare it to threshold values. Monitoring chemical contamination in the North Sea is also regulated by the OSPAR Regional Sea Convention. The OSPAR convention was adopted in 1992 to protect the marine environment of the North-East Atlantic and to guide international cooperation for the protection of the marine environment (WISE Marine, 2023). OSPAR defined primary contaminants as chemical compounds that may impose a risk on the marine environment and should thus be determined on a yearly basis (OSPAR commission, 2014). Primary contaminants consist of the sum of nine PAHs (anthracene, benzo[a]anthracene, benzo[ghi]perylene, benzo[a]pyrene, chrysene, fluoranthene, indeno[1,2,3-cd]pyrene, pyrene, phenanthrene), the sum of seven PCBs (CB28, CB52, CB101, CB118, CB138, CB153, and CB180) and eight trace metals (arsenic (As), cadmium (Cd), copper (Cu), chromium (Cr), mercury (Hg), nickel (Ni), lead (Pb) and zinc (Zn)). Secondary contaminants contain other PCBs, pesticides, anti-fouling agents, hydrocarbons, polychlorinated dibenzodioxins, polychlorinated dibenzofurans, etc. and should only be determined at sampling locations if a historical or active contamination source has been identified nearby (OSPAR commission, 2014).

Concentrations in sediments are in equilibrium with the marine environment. However, due to the hydrophobic character of PAHs, PCBs and metals, these contaminants tend to adsorb on sediment particles, resulting in a higher concentration in sediments compared to the aqueous environment (Nikolaou et al., 2009). The bioaccumulation of these compounds does not only depend on the environmental concentration but also on the bioavailability (Ruus et al., 2005). The mechanisms behind bioaccumulation and bioavailability can be species-dependent and are affected by physiological, biochemical, geochemical and ecological effects (Ruus et al., 2005). To determine contaminants in biomarkers, both vertebrates and invertebrates should be considered to have a more profound risk assessment of the marine ecosystem (Den Besten, 1998). Certain species may have mechanisms to tackle the uptake of contaminants. For example, they may have the ability to transform PAHs into hydrophilic metabolites, resulting in only trace amounts of PAHs detectable in their tissue (Vuorinen et al., 2006). This may thus result in a different contamination pattern or bioaccumulation. According to Net et al. (2015), PCBs were detected in the fish species *Sarotheroron melanotheron* and the bivalve *Perna perna*, whereas the concentration for PCBs remained below the detection limit in other species from the same region such as the fish species *Sarinella aurita* or the shrimp species *Penaeus Kerathurus*. Species like *Asterias rubens*, *Liocarcinus holsatus* and *Crangon crangon* are predators feeding themselves with other crustacea, molluscs, etc. (Choy, 1986; Den Besten et al., 2001b; Pihl and Rosenberg, 1984), which may result in the biomagnification of contaminants. Overall, determining the contamination in biomarkers can offer (early) warning signals when specific contaminants start to affect marine life (Den Besten, 1998). However, a good assessment of the environment requires the determination of contaminants in both biota and sediments (Chapman et al., 1996).

In this paper, the concentration of 11 PAHs (anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(ghi)perylene, benzo(k)fluoranthene, chrysene, fluoranthene, indeno(1,2,3-cd)pyrene, phenanthrene, pyrene), seven PCBs (CB28, CB52, CB101, CB118, CB138, CB153, CB180) and six metals (Cd, Cr, Cu, Hg, Pb, Zn) is assessed in *Asterias rubens*, *Liocarcinus holsatus* and *Crangon crangon* at five

disposal sites (LNP, Br&WOO, Br&WZE, Br&WS2, and Br&WS1), and reference locations in the Belgian part of the North Sea (BPNS) in the period 2012–2022. Contamination in biota is compared between disposal sites and reference locations, but also between disposal sites associated with an industrial port or a recreational harbour. Finally, the effect of bioaccumulation is studied in the long term and compared to the concentration measured in sediments.

2. Materials and methods

2.1. Sampling location

Sampling occurs at five disposal sites (LNP, Br&WOO, Br&WZE, Br&WS2, and Br&WS1) and reference locations (ft120, ft230, ft330, ft140tris) in the Belgian part of the North Sea (BPNS) (see Fig. 1). Disposal site LNP is located close to the harbour of Nieuwpoort, has an area of 1.77 km² and is only used for the disposal of dredged material originating from the harbour of Nieuwpoort. In 2020, LNP had a disposal rate of $0.23 \cdot 10^6$ ton dry matter per year. Br&WOO is located close to the port of Oostende and has an area of 1.77 km² with a disposal rate of $0.74 \cdot 10^6$ ton dry matter per year (2020). It is only used for the disposal of dredged material originating from the port of Oostende. For the port of Zeebrugge, three disposal sites are defined: Br&WZE, Br&WS2, and Br&WS1. Br&WZE is only used for the disposal of dredged

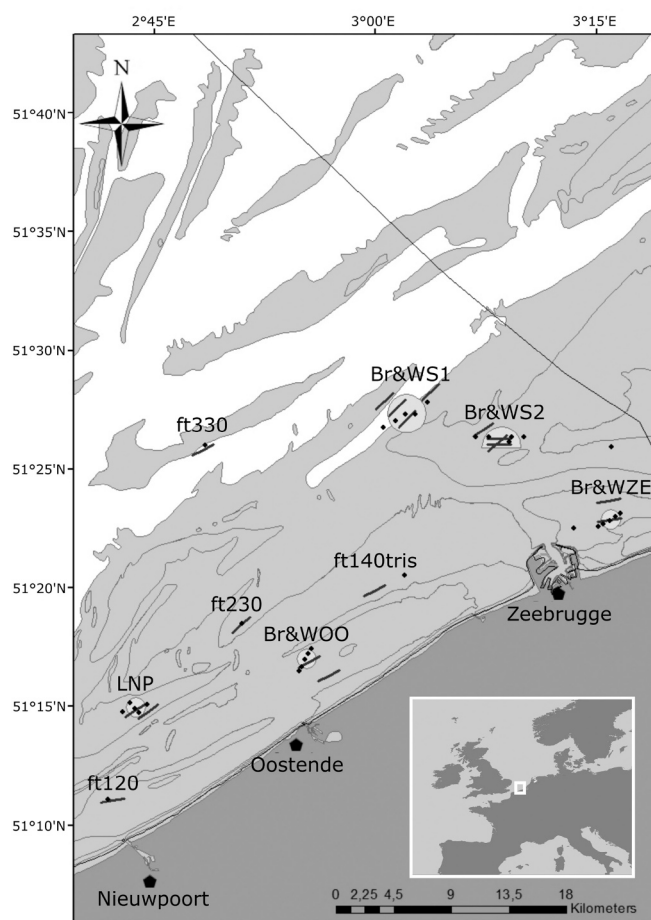


Fig. 1. Overview of the five disposal sites (LNP, Br&WOO, Br&WZE, Br&WS2, Br&WS1), reference location (ft120, ft140tris, ft230, ft330), and the three major ports and harbours (Nieuwpoort, Oostende, Zeebrugge) in the BPNS. Grabs are shown as • and fish tracks are shown with a line with a length of 1 nautical mile. Additionally, the seafloor depth is illustrated in grey (undep) and white (deep), and the outlines of the sandbanks are added with a thin black line.

material originating from the port of Zeebrugge, has an area of 1.77 km² and a disposal rate of 3.7 • 10⁶ ton dry matter per year (2020). Both Br&WS2 and Br&WS1 are also used for the disposal of dredged materials from the shipping channel connecting the port of Zeebrugge with the Western Scheldt. Br&WS2 and Br&WS1 have an area of 0.88 and 7.07 km² and a disposal rate of 1.4 • 10⁶ and 6.4 • 10⁶ ton dry matter per year (2020), respectively.

2.2. Sampling method and preservation

Three target species (*Asterias rubens*, *Liocarcinus holsatus* and *Crangon crangon*) were collected using a beam trawl of 8 m and a net with a cod-end mesh size of 40 mm (stretched) in the period 2012–2022 in February/March and September/October. Each fish track was one nautical mile (nm) long and the start and end coordinates of each fish track are reported in the Supplementary Material. The target species were collected from the catch and deep frozen (–20 °C) aboard the research vessel (RV) Belgica. Depending on the catch, the three target species were not always omnipresent at each location. Therefore, the concentration of the different contaminants could not always be determined for each species at each location. Due to an increased amount of litter accumulating at Br&WZE (Vanavermaete et al., 2023b), the collection of biota using a beam trawl became difficult resulting in no biota data after 2018 for that area.

2.3. Chemical analysis of PAHs

The concentration of 11 PAHs (anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(ghi)perylene, benzo(k)fluoranthene, chrysene, fluoranthene, indeno(1,2,3-cd)pyrene, phenanthrene, pyrene) was determined in the three target species. Extraction was done by freeze-drying (Labconco Freezone 6 L) tissue samples followed by pressurised liquid extraction (PLE) with an ASE350 (Dionex). In each extraction cell (22 mL), florisil (Merck, 0.150–0.250 mm) and diatomaceous earth (Celite 545, Sigma-Aldrich) were added together with 20 µL solution with recovery standards (200 µg mL^{–1} per compound) for quantification (naphthalene-d8, acenaphthene-d10, anthracene-d10, pyrene-d10, benzo(a)-pyrene-d12, and indeno(1,2,3-cd)pyrene-d12). An extraction program using a hexane:acetone 3:1 mixture over three extraction cycles at 100 °C with 5 min static time, 5 min heat time and 60 % rinse volume was used. The extraction cells were purged using nitrogen gas for 60 s. The extract was evaporated to exactly 1 mL using a Turbovap II evaporator (Zymark) at 30 °C, fractionated using a glass column (10 mm internal diameter) filled with 2 g of aluminium oxide (Aluminium oxide 90 active basic, Merck) and deactivated with 10 % of deionized type 1 water. Next, the extract was eluted with 15 mL of hexane and again evaporated to 1 mL and fractionated using a glass column (10 mm internal diameter) filled with 1 g of silica (Silica gel 60, Merck) that was conditioned with 10 mL hexane. The extract was eluted with 28 mL of hexane and evaporated and reconstituted into 0.5 mL isooctane. Before injection, an internal standard of 20 µL of a 200 µg mL^{–1} chrysene-d12 solution in isooctane was added to the vial. Next, 5 µL of sample was injected on an Agilent 7890 A GC equipped with a PTV-injector containing a glass bead liner (Gerstel, 6495-U) and temperature programmed from 80 to 320 °C. Separation was achieved on a select PAH column (Agilent, 30 m, 0.25 mm, 0.25 µm) with helium (Air Liquide, Alphagaz 2, ≥ 99.9999 %) as carrier gas and operated at a constant flow rate of approximately 2.2 mL min^{–1}. The retention time was kept constant using retention time locking. The oven program started at 50 °C and was kept constant for 1.05 min followed by an increase of 40 °C min^{–1} to 70 °C. Next, the temperature was increased with a rate of 35.5 °C min^{–1} to 180 °C followed by an increase of 3.55 °C min^{–1} to 230 °C which was held for 13.8 min. Finally, the temperature was increased to 280 °C at a rate of 25.36 °C min^{–1}, held for 13.8 min before increasing the temperature to 325 °C at 15.22 °C min^{–1} which was held for 2 min. Detection was done with an Agilent 5975C MS-detector with

electron ionization in selected ion mode with the transfer line temperature at 340 °C, ion source temperature at 300 °C, and quadrupole temperature at 150 °C.

2.4. Chemical analysis of PCBs

The concentration of seven PCBs (CB28, CB52, CB101, CB118, CB138, CB153 and CB180) was determined in the three target species. Between 2012 and 2020, PCBs have been determined by taking 80–100 mg of lipid and purifying it with 2 × 25 mL of hexane on a glass column (10 mm internal diameter) filled with 10 g of aluminium oxide (Merck, Aluminium oxide 90 active basic) and deactivated with 5 % of deionized type 1 water. The extract was evaporated to 1 mL using a Turbovap II evaporator (Zymark) before being eluted with 12 mL of hexane (Merck, Suprasolv, ≥ 98.0 %) on a glass column filled with 3 g of silicon oxide (Merck, Silica gel 60) that was deactivated with 5 % of deionized type 1 water. The extract was evaporated and reconstituted into 1 mL of isooctane (Merck, Lichrosolv, ≥ 99.0 %). 3.5 µL was injected on a Varian 450–GC with a split/splitless injector at 210 °C. Separation was done using an HT-8 column (SGE, 50 m, 0.22 mm, 0.25 µm) and hydrogen (Parker hydrogen generator) as carrier gas with a flow rate of 0.8 mL min^{–1}. The temperature program started at 90 °C and increased linearly to 285 °C over 54 min. Detection was done using an ECD-detector at 300 °C with 25 mL min^{–1} 95 % Ar–5 % methane as make-up gas. Tetrachloronaphthalene (TCN) was used as an internal standard and for calibration. Dilutions were made out of 100 ppm solutions of CB28, CB52, CB101, CB118, CB138, CB153 and CB180 in isooctane purchased from LGC standards.

Since 2020, PCBs have been analysed using a similar extraction method as described for PAHs. A 22 mL cell was filled with freeze-dried sample, florisil (Merck, 0.150–0.250 mm) and diatomaceous earth (Celite 545, Sigma-Aldrich). Isotope labelled C13 PCB congener mixture (CB28, CB52, CB101, CB118, C138, CB153 and CB180) was added as calibration standard to each cell. Extraction was done using pressurised liquid extraction (PLE) with an ASE350 (Dionex) using the same extraction program as described in Subsection 2.3. After evaporating the extract to 1 mL with a Turbovap II evaporator (Zymark), the extract was refracted as described earlier (see Subsection 2.3). Before splitless injection, TCN was added as an additional internal standard. PCBs were analysed using GC-MS² (TRACE 1300 with TSQ 9000; Thermo Fisher Scientific). A Rxi-5Sil MS Cap column of 30 m was used (0.25 mm ID × 0.25 µm) for separation. The temperature program started at 50 °C for 2 min before increasing to 180 °C at a rate of 28 °C min^{–1}. Next, the temperature was increased to 285 °C at a rate of 2 °C min^{–1} before reaching 335 °C at a rate of 25 °C min^{–1} where the temperature was held for 8 min. Detection was done with a TSQ 9000 MS-detector with an advanced electron ionization in selected ion mode with the transfer line temperature at 320 °C, ion source temperature at 300 °C.

2.5. Chemical analysis of metals

Biota samples of the three target species were homogenized prior to the analysis of six metals (Hg, Pb, Cd, Cr, Cu and Zn), using an analytical mixer (Heidolph Silent Crusher). Total Hg was determined after dry combustion with an oxygen stream followed by Au-adsorption and AAS-quantitation using a Direct Mercury Analysis (DMA-80, Milestone). For analysis of the other metals, 500 mg of the homogenized samples were digested (closed microwave digestion, MARS 6, CEM) in a 1:1 (v:v) mixture of nitric acid and deionized type 1 water (total volume 8 mL) at 180 °C for 30 min. The resulting digested solutions were diluted 4.3 times for measurement of Cu and Zn by ICP-OES (Varian 720, Agilent Technologies) or 10.3 times for measurement of Cr, Cd, and Pb by ICP-MS (Varian820, Analytik Jena & iCap RQ, Thermo Scientific).

2.6. Determining the amount of dry matter, lipid matter and total carbon

To determine the content of dry matter, the sample was mixed with 1 to 2 g of sand (Merck, Sea Sand extra pure) per g of sample and dried at 105 °C for at least 4 h. Since September 2013, samples have been dried using a freeze-dryer (Labconco Freezone 6 L). The dry matter content was then measured gravimetrically.

From 2012 to 2015, the total lipid weight was determined according to the Bligh and Dyer method (SWF, 1963). Starting with 10 g of wet sample, 20 mL of methanol (Biosolve, ≥99.9 %) and 2 × 10 mL of chloroform (Merck, Suprasolv, ≥99.8 %) was added. After adding 10 mL of deionized type 1 water, the mixture was filtered over a Whatman quantitative filter (GE healthcare, Grade 589/3). The filter content was mixed with 30 mL of chloroform and filtered over a second Whatman quantitative filter. After 2 h, the solvent phase was separated and evaporated using a rotavapor at 70 °C (Büchi, R-114). The total lipid was further dried in an oven at 50 °C for 30 min before determining the total lipid weight gravimetrically.

Since 2015, the total lipid weight is determined based on Smedes (1999). Starting with 5 g of wet sample, 16 mL of isopropanol and 20 mL of cyclohexane were added. After mixing (Ultra Turrax T25, IKA) the sample, 15 mL of deionized type 1 water was added and the sample was thoroughly mixed again before centrifugation at 2000 rpm for 5 min. The supernatant was transferred in an Erlenmeyer and 20 mL of a 13 % isopropanol mixture in cyclohexane was added. Again, the sample was thoroughly mixed and centrifuged at 2000 rpm for 5 min. The supernatant was once again transferred to an Erlenmeyer and solvent was evaporated using a rotary evaporator at 70 °C (Büchi, R-114) and further dried in an oven at 50 °C for 30 min before determining the total amount of lipids gravimetrically.

The total organic carbon (TOC) from a sediment sample was analysed using a mixture of dichromate-sulphuric acid followed by titration with 0.2 N Mohr's salt and *N*-phenylanthranilic acid as indicator.

2.7. Determining the contamination in sediments

The concentration of PAHs, PCBs and metals (including aluminium) was also determined in sediments. A complete overview of the analytical procedure is reported in Vanavermaete et al. (2023a).

2.8. Quality assessment and quality control

For each analysis sequence, positive control samples and procedure blanks were analysed. Proficiency testing was done for each analysis (biota and sediment matrix) by yearly participation at the Quasimeme interlaboratory exercises (Quasimeme, Wageningen, The Netherlands), focused on marine samples. The analysis of PAHs, TOC and total lipids are ISO/IEC 17025 accredited. Quantification limits for PAHs, PCBs and metals are given in the Supplementary Material (see Tables S2–5).

2.9. Statistical analysis

The statistical analysis was performed in R (version 4.1.2). In a first step, data containing the concentrations of PAHs, PCBs and metals as well as the dry weight and total lipid weight in *Asterias rubens*, *Liocarcinus holsatus* and *Crangon crangon* between 2012 and 2022 were collected. Concentrations below the detection limit were replaced by the detection limit (see Supplementary Material). Originally, PAHs were expressed in dry weight, PCBs in lipid weight and metals in wet weight. Therefore, the concentration was recalculated using the dry and lipid weight of each organism. If the dry or lipid weight of an organism was unknown, the measured concentrations were omitted from the study. All results were calculated on dry weight concentrations, except for the long-term trends that use wet weight (see further). Before analysis, measured concentrations as well as lipid and dry weight that exceeded more than three times the standard deviation from their mean value

were considered outliers and removed from the study.

To compare the bioaccumulation of chemical contaminants in their environment, the concentration measured in sediment grabs was assessed (Vanavermaete et al., 2023a). Because the latter may be influenced by the physical composition of the soil, a normalization is needed (ICES, 2009). PAHs and PCBs were normalized to a TOC value of 2.5 % whereas metals were normalized to an aluminium (Al) concentration of 5.8 % as reported in Vanavermaete et al. (2023a). After normalization, grabs that were sampled in the same area were pooled together (using the log mean) and linked to the current dataset.

The accumulation of sediments in biota was studied by calculating the biota sediment accumulation factor (BSAF). For a specific species, the BSAF of the *i*-th compound at the *j*-th location was calculated as:

$$BSAF_{ij} = \frac{c_{wet,ij}}{c_{ss,ij}} \times 100, \quad (1)$$

with $c_{wet,ij}$ the mean wet weight concentration of the *i*-th compound measured in this species at the *j*-th location and $c_{ss,ij}$ the mean normalized concentration of the *i*-th compound measured in sediment samples at the *j*-th location. This indicator, which was calculated in the period 2018–2022, represents the relative accumulation of a specific compound in a specific species compared to its environment.

Differences in concentrations between areas were analysed using ANOVA and the *aov* function in R. A log transformation of the data was necessary to fulfill the requirements of ANOVA, which was checked and validated visually using a QQ-plot and the plot of residuals versus fitted values. The log mean difference between every two areas was then calculated using the Tukey honest significance test (TukeyHSD). Because data was not available for each species studied in each area, results were only included in the ANOVA if data was available for at least three years. This was important to calculate a stable and reliable log-mean concentration. The overall contamination pattern was assessed using principal component analysis (PCA) and the *prcomp* function in R to reduce the measured concentrations of PAHs, PCBs and metals into two principal components (PC). Before analysis, the data was rescaled between 0 and 1.

Long-term trends in three target species were analysed using mixed effects models (GLME). The location was considered as a random effect, while the analysis method was considered as a fixed effect. Starting from this design, potential fixed effects were added to the model. Seven variables were considered: sampling year, season, area (LNP, Br&WOO, Br&WZE, Br&WS2, Br&WS1 and R), dry weight, lipid weight, TOC and (normalized) concentration in sediments. Additionally, three interaction terms were considered combining the dry weight, lipid weight and TOC with the concentration measured in sediments. In total 279 models were evaluated using the *aictab* function. The model with the lowest AICc that reached convergence was selected and used for further discussion. If a variable had a variance inflation factor (VIF) higher than 5, then the model was deemed invalid (dependent variables) and the next best model was selected instead. The VIF of the variables that were included in the interaction term were not taken into account as long as the interaction term was significant ($p < 0.05$).

3. Results and discussion

Asterias rubens has often been used as a biomarker to measure the effects of stressors such as chemical contamination on the benthic population (Den Besten et al., 2001b,a; Canty, 2009). To avoid repetition, the study is focused on the chemical contamination in *Asterias rubens* and the main differences in contamination observed in both *Liocarcinus holsatus* and *Crangon crangon* are highlighted at the end of this section.

3.1. Short-term bioaccumulation of contaminants in *Asterias rubens*

The bioaccumulation of PAHs, PCBs and metals in *Asterias rubens* was studied over a short-term period of five years (2018–2022). The overall chemical pattern at each disposal site was studied using PCA, reducing the measured concentrations of PAHs, PCBs and metals into two principal components (PCs). Each location (haul) was labelled according to the area (LNP, Br&WOO, Br&WZE, Br&WS2, Br&WS1 and R) (Fig. 2). The difference in the log concentration for each compound at each location (disposal sites vs reference locations) was also compared using ANOVA. The difference in concentration for each disposal site at which at least three years of data was available is illustrated in Fig. 3, where a concentration of 1 indicates a 100 % higher concentration than observed at reference locations.

The first principal component explains approximately 84–92 % of the variance and is correlated to the total amount of contamination. Locations situated in the left side of the PCA plot contain higher concentrations of PAHs, PCBs or metals than locations situated in the right side. For PCBs and to a lesser extent for PAHs, a distinction can be made between dredged material associated with a recreational harbour (LNP) and an industrial port (Br&WOO and Br&WZE). As observed in sediment samples, the highest PCB contaminations are observed at Br&WZE, followed by Br&WOO and LNP, which cluster together with reference locations (Vanavermaete et al., 2023a). This is also observed in Fig. 3, where relatively higher concentrations of PAHs and PCBs are observed for Br&WOO compared to LNP (no data was available for Br&WZE). According to the Tukey honest significant test, only CB101–CB153 had a significantly higher ($p < 0.05$) concentration at Br&WS1 and/or Br&WOO compared to reference locations.

The contamination pattern of metals differs. Br&WOO clusters together with LNP and reference locations, whereas a higher contamination is observed at Br&WZE and Br&WS1–2 (Fig. 2C). The Scheldt estuary could be a possible source. Gaulier et al. (2021) reported an overall overview of chemical contamination in the Scheldt estuary based on different studies published in the period 1978–2019. Especially close to the port of Antwerp, the contamination of metals was higher than measured in the marine sediment. Because Br&WZE is situated in a sedimentation area close to the outlet of the Scheldt, an accumulation may be expected. Additionally, in a study of 2006, high concentrations

of metals and PCBs in *Asterias rubens* were observed at the outlet of the Scheldt estuary (Danis et al., 2006). This could explain why similar effects for metals and PCBs are observed in the overall PCA plot (panel D, Fig. 2). Taking into account that PCBs were banned in the '80s (Faroon et al., 2003), a decrease in their concentration is expected. This was generally observed in the BPNS, except for the area near the Scheldt estuary (Everaert et al., 2014). Part of the dredged material that is disposed of at Br&WS1–2 originates from the Scheldt estuary and nearby shipping lanes. As a consequence, a similar contamination is observed on the PCA plot as for Br&WZE.

According to the PCA plot of metals (panel C), the first PC expresses the overall contamination of metals, whereas the second PC differentiates between the different metals. Sampling points situated at Br&WS1 have a relatively higher concentration of Pb. According to Belgische Staat (2018), the Scheldt estuary is characterised by a high Cu/Zn contamination. The increased Pb concentration observed at its dredge disposal may thus indicate that other effects or sources are also at play.

To quantify the relative difference in the concentration measured in sediment and the concentration measured in *Asterias rubens*, the biota sediment accumulation factor (BSAF) was calculated (Table 1). An overall low BSAF is observed for PAHs. Only phenanthrene and pyrene exceeds the value of 1 %. In an exposure study of PAHs in *Asterias rubens*, it was shown that after 24 days, an overall decrease in the level of PAHs in the tissue was observed except for pyrene, phenanthrene and chrysene (Joly-Turquin et al., 2009). Although this is not observed for chrysene and pyrene, this may explain the higher BSAF observed for phenanthrene.

For PCBs, a BSAF higher than 100 % is observed, indicating that a higher accumulation in *Asterias rubens* was observed compared to sediment samples. This is also observed for CB153 in Danis et al. (2004) where concentrations in sediment (dry weight) and *Asterias rubens* (lipid weight) are available. Additionally, the reported values of low-impact stations ranged between 131 and 191 ng g⁻¹ lipid weight for CB153, which is slightly higher than the concentrations measured in this study (77–154 ng g⁻¹ lipid weight), possibly indicating a long-term decrease.

A higher BSAF is observed for both Zn (21 %) and Hg (28 %). Zinc is often used in anti-fouling paints (Lagerström et al., 2018). The measured contamination of Zn (but also Cu) can partly be related to the ship traffic volume and both metals are also correlated to the effluent of the Scheldt

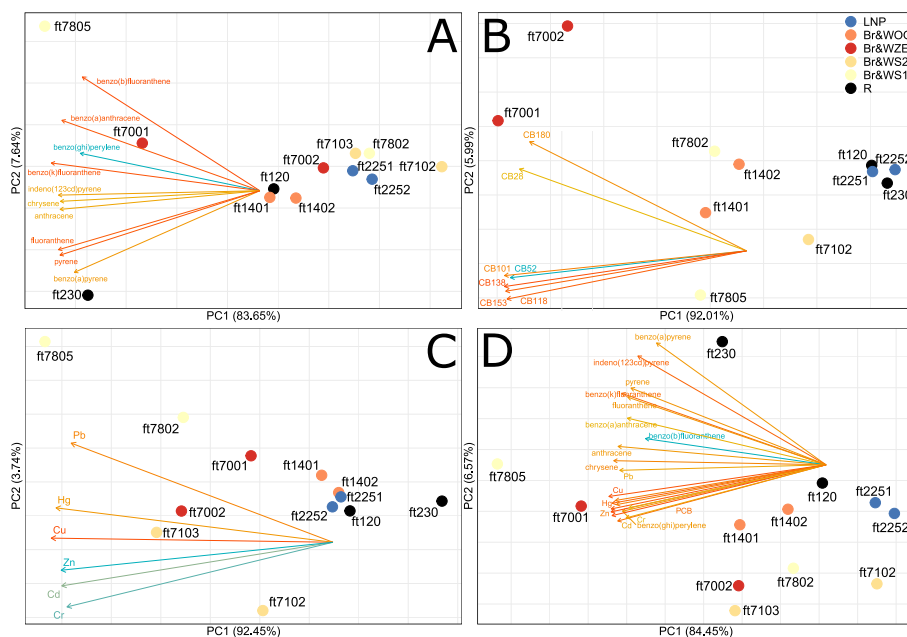


Fig. 2. Overview of the contamination pattern of PAHs (panel A), PCBs (panel B), metals (panel C) and all contaminants (panel D) in *Asterias rubens* at each disposal site (LNP, Br&WOO, Br&WZE, Br&WS2, Br&WS1) and reference locations (R) using PCA in the period 2018–2022.

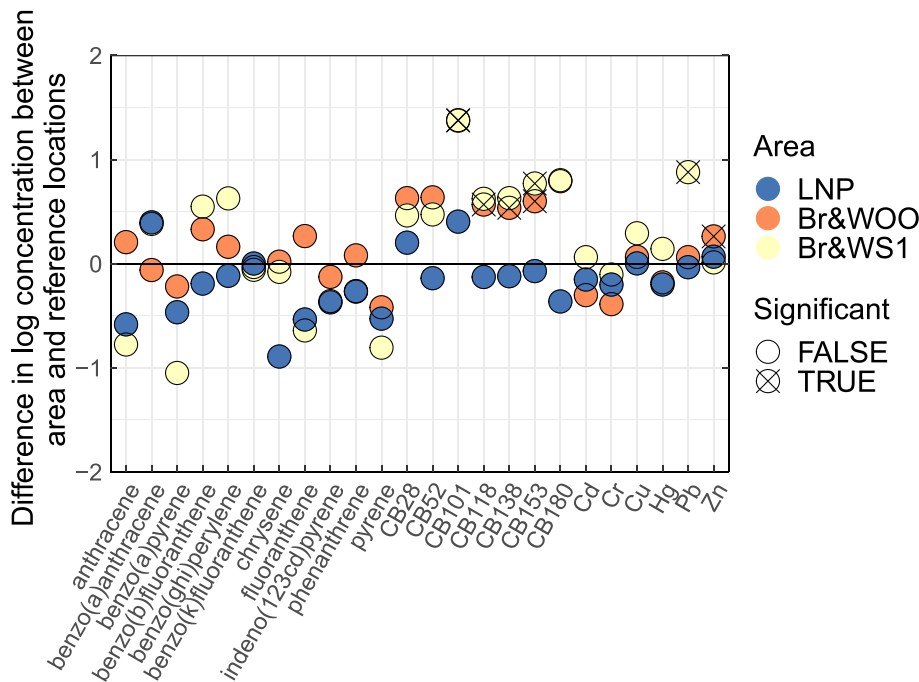


Fig. 3. The difference in the mean log concentration in *Asterias rubens* for each contaminant (PAHs, PCBs and metals) between disposal sites and reference locations using the Tukey honest significance test in the period 2018–2022. Significant results ($p<0.05$) are shown with ×.

Table 1
The biota sediment accumulation factor (BSAF) measured in *Asterias rubens* and sediment, calculated according to Eq. (1), and expressed in percentage for the period 2018–2022. A value > 100 indicates that a higher concentration was measured in *Asterias rubens* than in the sediment in that same area.

	LNP [%]	Br&WOO [%]	Br&WZE [%]	Br&WS2 [%]	Br&WS1 [%]	R [%]
Anthracene	0.19	0.37	0.48	0.13	0.22	0.37
Benzo(a)anthracene	0.23	0.13	0.37	0.11	0.36	0.18
Benzo(a)pyrene	0.38	0.49	0.28	0.19	0.36	0.74
Benzo(b)fluoranthene	0.06	0.10	0.23	0.07	0.21	0.08
Benzo(ghi)perylene	0.16	0.20	0.30	0.33	0.51	0.20
Benzo(k)fluoranthene	0.28	0.24	0.42	0.12	0.39	0.30
Chrysene	0.12	0.29	0.33	0.20	0.44	0.34
Fluoranthene	0.34	0.66	0.91	0.28	0.47	0.66
Indeno(123 cd)pyrene	0.12	0.15	0.10	0.04	0.20	0.19
Phenanthrene	1.87	2.56	3.31	2.19	2.85	2.89
Pyrene	0.51	0.51	1.01	0.29	0.59	1.00
CB28	14.2	22.1	40.9	7.39	24.7	12.1
CB52	44.7	109	114	103	92.5	49.8
CB101	59.5	164	281	168	179	38.9
CB118	54.0	122	220	133	156	66.5
CB138	92.6	196	333	195	235	109
CB153	133	249	377	294	367	132
CB180	13.0	28.0	98.8	4.17	32.7	14.4
Cd	0.02	0.02	0.01	0.03	0.03	0.03
Cr	1.54	1.41	1.90	2.31	1.48	1.94
Cu	7.37	9.65	11.5	11.1	10.4	8.28
Hg	25.7	22.9	31.9	22.8	45.2	29.8
Pb	0.60	0.72	0.86	0.53	1.68	0.67
Zn	24.7	37.6	44.3	24.7	23.1	24.1

estuary (Belgische Staat, 2018). In Fig. 3, a higher concentration of Zn can also be observed at an industrial disposal site (Br&WOO), whereas for Cu, the difference is not significant. In an earlier study, higher Zn concentrations with relatively lower Cu concentrations were also observed in *Asterias rubens* at Br&WZE (referred to as Knokke) (Danis et al., 2006). Copper is better regulated in *Asterias rubens*, resulting in a lower BSAF (7 %) (Coteur et al., 2003). According to Coteur et al. (2003), an accumulation of Cd and Pb was also observed in *Asterias rubens*. This is, however, less clearly observed in this study with only an BSAF of 0.02 % (Cd) and 0.8 % (Pb).

3.2. Long-term bioaccumulation in *Asterias rubens*

In the previous sections, the difference in contamination between the different disposal sites and reference locations was unravelled. To study trends in bioaccumulation of PAHs, PCBs and metals in the long term (2012–2022), a linear mixed effects model was used. For each compound, 279 models were constructed and based on the AICc, the most proficient model was selected. An overview of the most proficient models is shown in Table 2.

Except for benzo(k)perylene, phenanthrene and pyrene, at least one non-singular model was constructed for each PAH. The R^2 varies

Table 2

Overview of the mean estimated value for each variable used to model the long-term concentration of PAHs, PCBs and metals in *Asteria rubens* in the period 2012–2022. If no value is provided, that variable was removed to avoid singularity. Significant variables are marked with. (p<0.10), * (p<0.05), ** (p<0.01), *** (p<0.001).

Compound	Year	Season	Location					Dry weight	Lipid weight	TOC	Sed	Lipid × Sed	Dry × Sed	TOC × Sed	Correlation R ²
			LNP	Br&WOO	Br&WZE	Br&WS2	Br&WS1								
Anthracene	4e-04 (***)	−0.0027 (***)						0.001			0.02 (***)				0.41
benzo(a)anthracene		−0.0021									0.02 (**)				0.29
benzo(a)pyrene		−0.0125 (**)							0.0085	−0.0012	−0.0244			−0.02	0.13
benzo(b)fluoranthene									−0.0024		0.02 (**)	−0.0372 (**)			0.25
benzo(ghi)perylene											0.01 (***)				0.21
benzo(k)fluoranthene															NA
chrysene		−0.0052 (***)							0.0033		0.0094				0.23
fluoranthene	0.0026 (**)	−0.0188 (***)									0.02				0.24
indeno(1,2,3-cd)pyrene	0.0016 (***)		−0.0019							−0.0016					0.3
phenanthrene															NA
pyrene															NA
CB28								−0.0012	0.53 (***)		0.34 (***)	0.88 (***)			0.74
CB52	−0.0087 (***)							−3.2267	7.5	−0.005	6.7	12.5	−5.3688		0.41
CB101								−0.0057	0.06 (***)		2.3				0.42
CB118	−0.0036 (**)	0.0092						0.22	0.04 (***)	0.54 (**)	0.13		0.38	0.91 (**)	0.54
CB138	−0.0068 (**)								0.78	0.91 (**)	−0.9614	1.2		1.5 (**)	0.52
CB153		0.04						−0.0292 (**)	4.6 (**)	0.99	3.1	7.6 (**)		1.7	0.5
CB180	−0.0013							−0.4959	2.1 (**)		1.5	3.5 (**)	−0.8202		0.35
Cd	2e-04 (***)	−0.0013 (***)						7e-04 (***)			−2e-04				0.18
Cr		−0.0227						−0.0196		−0.0309 (***)	−0.1362 (**)		0.12 (**)	0.1 (***)	0.21
Cu		0.03 (***)							−0.1069 (**)	−0.0084 (***)	0.13	−0.2388 (**)			0.28
Hg	−0.1224 (**)		−0.664					0.43 (**)	−0.8282 (**)	−0.2997					0.32
Pb								0.0067	−0.0105	0.0052 (**)	−0.058 (**)	−0.0537	0.03 (**)	0.03 (***)	0.16
Zn	−0.0692 (**)	−0.8516 (***)	−0.2914						0.42						0.37

between 0.13 for benzo(a)pyrene and 0.41 for anthracene. These correlations are generally low, requiring some caution when interpreting results. Two significant effects are observed, a seasonal effect and an environmental effect (concentration in sediment). The negative seasonal effect indicates that PAHs tend to accumulate in *Asterias rubens* with higher concentrations in the spring compared to the fall. Compared to the early spring (March), a growth in the pyloric caeca is expected between April and September resulting in a higher lipid and protein content and a lower contamination level (Den Besten et al., 2001a). The seasonal effect in *Asterias rubens* was also observed in other studies (Den Besten et al., 2001b; Baumard et al., 1999). The environmental effect indicates a clear link with the contamination measured in the sediment where the higher concentrations are observed at Br&WOO, followed by LNP and Br&WS1.

The models constructed for PCBs had an R^2 ranging from 0.35 to 0.74. For at least three PCBs (CB52, CB118 and CB138) a decrease in concentration over time is observed. This was also observed in sediment samples and can be linked to the ban of PCBs in the '80s (Faroon et al., 2003). Den Besten et al. (2001a) reported a seasonal effect for PCBs, however, the season had no significant effect in our model. This effect is captured by the change in lipid weight. PCBs accumulate in the adipose (fat) tissue. In the fall, a higher lipid content is observed which allows for higher accumulations of PCBs, hence the positive effect on the lipid weight. The increased accumulation is further confirmed by a significant interaction term between the lipid weight and the concentration in sediment. The higher contamination of PCBs in *Asterias rubens* in the fall is, however, only observed when the concentration is expressed in wet weight. Compared to the spring, in the fall the lipid weight increases at a higher rate compared to the concentration of PCBs. Therefore, when the PCB concentration is expressed in lipid weight ($\mu\text{g}/\text{kg}$ lipid), higher PCB concentrations (per unit of lipid) are detected in the spring.

For metals, the R^2 varied between 0.16 and 0.37. Whereas the bioaccumulation of the different PAHs and PCBs is correlated with the same variables, this is not the case for the metals. A seasonal effect is only observed for four out of six metals with higher concentrations in the spring for Cd, Cr and Zn and lower concentrations for Cu compared to the fall. Both Cd and Zn can be transported in the atmosphere over a long range before deposition in the marine environment and are connected with human-induced atmospheric fallout, which peaked in the '70s (Jokinen et al., 2020). The distribution of metals and their seasonal variation differs depending on the body compartment under investigation. Temara et al. (1997) reported higher metal concentrations in the pre-spawning period (April) in the pyloric caeca, whereas in the digestive tract, Cu concentrations were higher during gametogenesis (January), Zn concentrations during the post-spawning period (August) and other metals during the pre-spawning period (April). Taking into account that the body mass of each compartment is also bound to seasonal variation, it becomes complex to predict the level of contaminants in *Asterias rubens*. This may thus explain why significant effects are observed for dry weight or lipid weight for one metal but not the other.

In sediment samples, metals are normalized to aluminium to correct for the clay content of the sediment (OSPAR, 2009). However, metals complexes with organic material present in the soil (Tam and Wong, 1996). On the one hand, higher organic concentration can go hand in hand with a higher complexation (metal dependant) of metals, and thus higher metal concentrations (Yang et al., 2010; Reddy et al., 2004). On the other hand, organic matter can act as a carrier for the transportation and distribution of metals along the land-sea continuum (Mannio et al., 1993; Jokinen et al., 2020). Additionally, when metals are bound on organic matter, they can become less bioavailable explaining why a significant negative effect is observed between the TOC and the concentration of metals like Pb, Cu and Cr in *Asterias rubens*. A seasonal effect is also observed but is different for each metal.

3.3. Short-term trend in *Crangon crangon* and *Liocarcinus holsatus*

The contamination pattern observed in *Liocarcinus holsatus* between dredge disposal sites linked to an industrial port and recreational harbours is less distinct compared to *Asterias rubens* (see Fig. S1). The lowest contamination is observed in LNP (right side of the PCA) and the highest in Br&WZE (left side of the PCA plot) for PAHs, PCBs and metals. However, no further distinction can be made between the other dredge disposal sites and the reference location. For *Crangon crangon*, no pattern between both types of ports/harbours can be observed (see Fig. S2).

For *Liocarcinus holsatus*, the highest concentrations of individual contaminants can still be observed in Br&WOO, although these are not significantly ($p < 0.05$) higher than the concentrations observed at reference locations (see Fig. S3). For *Crangon crangon*, there is no clear link between the measured concentrations and those at the disposal sites (see Fig. S4).

The contamination in *Asterias rubens* and to a lesser extent *Liocarcinus holsatus* follow the same trend as observed in sediment samples, but differs from the one observed for *Crangon crangon*. This was also observed in the study of Hilvarsson et al. (2009), where a lower bioaccumulation of medetomidine in *Crangon crangon* compared to *Mytilus edulis* was measured. This indicates that bioaccumulation differs between species and that in order to understand the effects of environmental contaminants on biota, different species should be analysed. Additionally, species like *Crangon crangon* (and *Mytilus edulis*) are part of the human food chain, and are thus also important to include in monitoring programs to assess the quality and safety of seafood.

3.4. Long-term trend in *Crangon crangon* and *Liocarcinus holsatus*

The concentrations measured in sediment are less correlated with the concentrations measured in *Liocarcinus holsatus* and *Crangon crangon*. The time trend for PCBs in both *Liocarcinus holsatus* and *Crangon crangon* is similar to the one observed in *Asterias rubens*. This is not always the case for PAHs and metals. In *Liocarcinus holsatus*, a negative time trend is observed for benzo(ghi)perylene, Cd and Pb, whereas a positive time trend is only observed for Cu. In the case of *Crangon crangon*, a negative time trend is only observed for Pyrene. Only for indeno(1,2,3-cd)pyrene, a positive time trend is observed in both *Asterias rubens* and *Crangon crangon*.

In *Asterias rubens*, a seasonal effect is observed for the PAH contamination and a correlation between the lipid weight and the PCB concentrations. This link is less present in *Liocarcinus holsatus* (see Table S6) and *Crangon crangon* (see Table S7). In the case of *Asterias rubens*, a greater difference in lipid weight (1.76 ± 0.35 vs 2.80 ± 0.62) and dry weight (27.1 ± 1.69 vs 30.8 ± 1.39) is observed between the spring and fall surveys. This is not the case for *Crangon crangon* and *Liocarcinus holsatus* where a lipid weight of respectively 5.65 ± 1.45 vs 5.80 ± 1.67 and 1.03 ± 0.12 vs 1.18 ± 0.23 is observed between the spring and fall surveys. The dry weight varies between 24.8 ± 1.57 and 23.2 ± 3.39 for *Liocarcinus holsatus* and 22.5 ± 0.68 and 23.5 ± 0.86 for *Crangon crangon* in the spring and fall surveys, respectively. This may indicate why the seasonal effect is less or not observed in both *Liocarcinus holsatus* and *Crangon crangon* compared to *Asterias rubens*.

3.5. Conclusion

Measuring the concentrations of PAHs, PCBs and metals in marine sediments and *Asterias rubens* in and around different dredge disposal sites of the Belgian Part of the North Sea revealed that dredge disposal can be a vector for chemical contaminants. At the level of individual contaminants, only four PCBs and two metals revealed a significantly higher concentration at specific dredge disposal sites. However, when combining all contaminant data in a PCA, a distinct contamination pattern can be seen between the different dredge disposal sites, clearly associated with the origin of the dredge. Higher contaminant

concentrations can be measured at dredge disposal sites receiving dredge from industrial ports compared to sites associated with a recreational harbour. Important factors influencing bioaccumulation of contaminants in *Asterias rubens* include lipid weight and TOC. PAH concentrations vary seasonally in *Asterias rubens*. PCB contamination is correlated with lipid weight, indicating that PCB concentrations accumulate in the adipose tissue. For metals, correlations with lipid weight or TOC were observed, depending on the metal ion considered.

In monitoring studies, selection of appropriate biomarkers should take into account the purpose of the monitoring scheme. Our results demonstrate that *Asterias rubens* is an excellent choice as biomarker if the main goal is to have a proxy for environmental sediment concentrations of PAHs, PCBs and metals. However, if the main aim is to assess risk for bio-accumulation in the food chain or even effects on humans by food consumption, more species should be considered to allow for a more complete assessment. When this is required, selection of alternative or additional biomarkers such as *Liocarcinus holsatus* or *Crangon crangon* is advised. The latter species have a clearly distinct contamination pattern compared to *Asterias rubens* and exhibit a smaller seasonal variation in dry and lipid weight. Therefore, monitoring of *Liocarcinus holsatus* and *Crangon crangon* is less dependent on specific timing, while for *Asterias rubens*, sampling should consistently occur within in the same time window.

CRedit authorship contribution statement

David Vanavermaete: Writing – original draft, Visualization, Formal analysis. **Bavo De Witte:** Writing – review & editing, Supervision, Conceptualization. **Kris Hostens:** Writing – review & editing, Supervision, Conceptualization. **Ann Ruttens:** Writing – review & editing, Formal analysis. **Nadia Waegeneers:** Writing – review & editing, Formal analysis. **Karliene Cheyns:** Writing – review & editing, Formal analysis. **Jenne Leus:** Formal analysis. **Bernard De Baets:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2024.117068>.

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